Подстановка $x = 2\cos\theta$ и многочлены Чебышёва

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Опр. The Chebyshev polynomials are defined by $T_n(\cos\theta) = \cos n\theta$ and $U_{n-1}(\cos\theta) = \frac{\sin n\theta}{\sin\theta}$; equivalently $T_0 = 1$, $T_1 = x$, $T_{n+1} = 2xT_n - T_{n-1}$. On $|x| > 1$ the same recurrence is realised by $T_n(\cosh t) = \cosh nt$.

Теорема 1 (экстремальное свойство, Чебышёв 1854) *Among all monic real polynomials of degree n the polynomial $2^{1-n}T_n$ has the smallest possible maximum modulus on $[-1,1]$, namely 2^{1-n}, and it is the unique such polynomial.*

Теорема 2 (Марков 1889, Бернштейн 1912) *If $\deg p \leqslant n$ and $|p| \leqslant 1$ on $[-1,1]$, then $|p'| \leqslant n^2$ on $[-1,1]$, with equality only for $\pm T_n$ at the endpoints; and $|p'(x)| \leqslant n/\sqrt{1-x^2}$ inside.*

Факт 1 The substitution $x = 2\cos\theta$ turns $x^2 - 2$ into $2\cos 2\theta$; hence the map $x \mapsto x^2 - 2$ acts on $[-2,2]$ as doubling of the angle, and ± 2, ± 1 are its only rational periodic points.

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1. Prove that T_n is a polynomial of degree n with leading coefficient 2^{n-1}, prove Theorem 1, and use it to show that $\max_{[-1,1]} |p| \geqslant 2^{1-n}$ for every monic p.

2. Prove Theorem 2 for the special case $p = T_n$, then deduce the general Bernstein inequality $\|T'\|_\infty \leqslant n\|T\|_\infty$ for trigonometric polynomials of degree n.

3. Let $a_1 = a$ and $a_{n+1} = a_n^2 - 2$. Describe the behaviour of the sequence for every real a, and prove that a_n is an integer for all sufficiently large n only when $a \in \{0, \pm 1, \pm 2\}$ or $a = t + t^{-1}$ with t an algebraic unit.

4. (Proposed problems) Let (T_n) be the Chebyshev polynomials and let (F_n) be the Fibonacci sequence, $F_0 = 0$, $F_1 = 1$. Prove that $T_n\left(\frac{i}{2}\right) = \frac{i^n L_n}{2}$ and $U_{n-1}\left(\frac{i}{2}\right) = i^{n-1} F_n$, where $L_n = F_{n-1} + F_{n+1}$; deduce Catalan's identity for Fibonacci numbers from the corresponding identity for T_n.

5. (IMC) Let $\theta > 0$ and let $\cosh t = \frac{e^t + e^{-t}}{2}$. Show that if $k \in \mathbb{N}$ and both $\cosh k\theta$ and $\cosh(k+1)\theta$ are rational, then so is $\cosh \theta$.